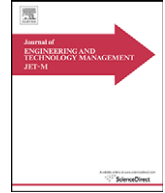




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The social structure of leadership and creativity in engineering design teams: An empirical analysis

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ABSTRACT

Creativity is essential for research and development efforts. Unfortunately, little is known about how the role of team leaders determines the team's creativity. Based on a sample of 39 engineering design teams in the space industry, this study examines the effects of leader position within different flows of communication on team creativity. The results indicate that the balance between holding a central or a peripheral position indeed determines the creativity of such teams. Specifically, very central and very peripheral positions of team leaders within the work-flow and awareness network hamper team creativity, whereas peripheral situated team leaders within the information network propel the creativity. In addition, team leaders stimulate creativity when they stay central in the external information network. In managerial terms, the analyses implicate that stimulating engineering design team creativity requires team leaders to smartly limit their involvement in communication and mainly act as gatekeepers to external sources of information.

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1. Introduction

As firms struggle to establish or maintain prosperity in turbulent and competitive environments, creativity becomes increasingly important (DeVanna and Tichy, 1990). Researchers widely agree that creativity is essential for the long-term survival of organizations, since it is the foundation of organizations' innovative potential (Badawy, 1993; Glynn, 1996; Oldham and Cummings, 1996; Scott

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